Practice workbook · Areas and perimeters of shapes made from sectors and triangles

Finding the area or perimeter of a region built from sectors, segments, semicircles and triangles, using (1/2)ab sin C and the cosine rule for straight edges.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-05

1. Question 4 · part whole · 7 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

4

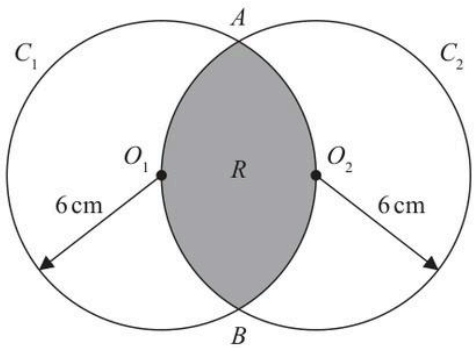


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows two circles, C_1 and C_2, each with a radius of 6 cm.

The centre of C_1 is O_1 such that O_1 lies on C_2

The centre of C_2 is O_2 such that O_2 lies on C_1

The circles intersect at the points A and B and enclose the region R, shown shaded in Figure 1

The area of region R is P cm^2

Find the exact value of P, giving your answer in the form $a\pi - b\sqrt{c}$ where a, b and c are integers.

(7)

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8

P 7 1 6 6 6 A 0 8 3 6

1. Question 4 · part whole · 7 marks

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Worked-example retry · Attempt part whole only.

Question 4 continued

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1. Question 4 · part whole · 7 marks

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Worked-example retry · Attempt part whole only.

Question 4 continued

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1. Question 4 · part whole · 7 marks

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Worked-example retry · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

1. Question 4 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 3 · part c · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

3

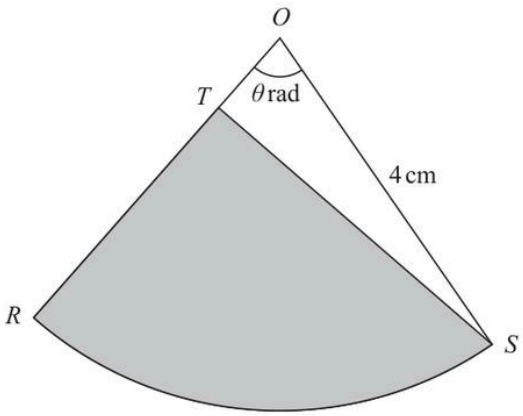


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector ORS of a circle with centre O and radius 4 cm.
The size of angle ROS is θ radians.

The area of sector ORS is 2π cm^2

(a) Find the exact value of θ

(2)

(b) Find the perimeter, in cm to 3 significant figures, of the sector ORS.

(2)

The point T lies on OR such that $OT : TR = 1 : 3$

The region shown shaded in Figure 1 is bounded by the line TR, the line TS and the arc RS of the sector.

The area of this region is A cm^2

(c) Find the exact value of A

(2)

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2. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 6 marks)

2. Question 3 · part c · 2 marks · Working and answer

Working and answer

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3. Question 9 · part a · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

9

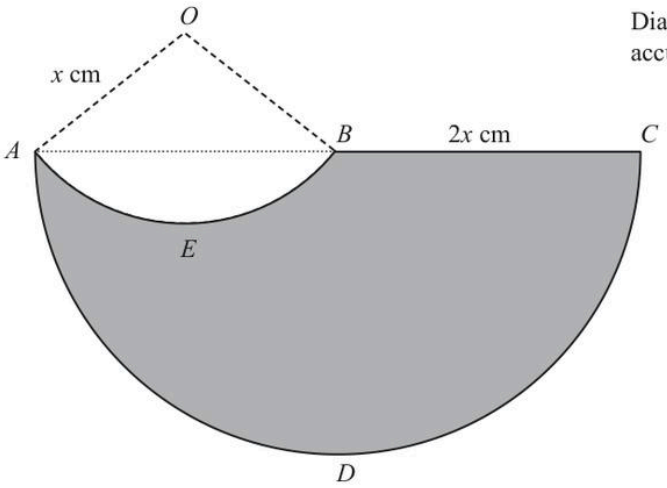


Diagram **NOT** accurately drawn

Figure 2

A logo, $AEBCD$, is shown shaded in Figure 2.

The straight line ABC is the diameter of the semicircle ADC
AEB is an arc of a circle with centre O
All angles are measured in radians.

- $BC = 2x$ cm
- $OA = OB = x$ cm
- length of arc $AEB = 1.8x$ cm

The perimeter of the logo is P

(a) Show that $P = ax(\pi + \pi \sin 0.9 + b)$ where a and b are constants to be found.

(7)

Given that $x = 10$ cm,

(b) find, in cm^2 to 3 significant figures, the area of the logo.

(6)

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3. Question 9 · part a · 7 marks · Working and answer

Working and answer

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4. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

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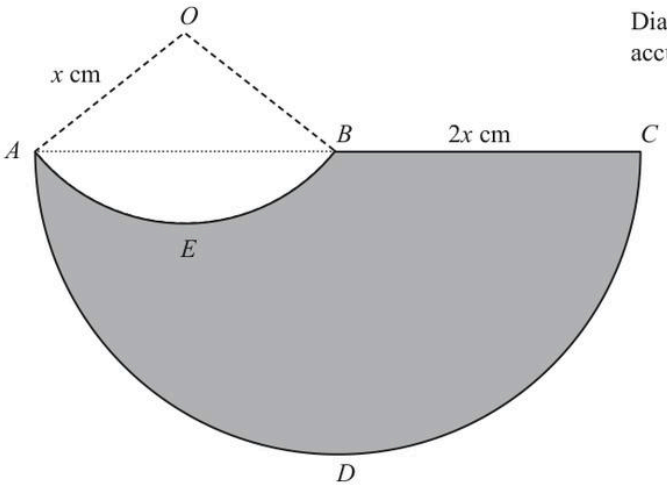


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4. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

Question 9 continued

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4. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

Question 9 continued

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4. Question 9 · part b · 6 marks

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Further practice · Attempt part b only.

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Question 9 continued

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(Total for Question 9 is 13 marks)

4. Question 9 · part b · 6 marks · Working and answer

Working and answer

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5. Question 3 · part b · 4 marks

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Further practice · Attempt part b only.

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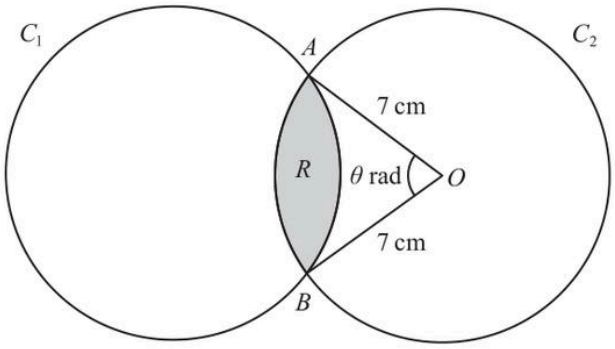


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows circle C_1 with radius 7 cm and circle C_2 with centre O and radius 7 cm

The circles C_1 and C_2 intersect at the point A and at the point B

The size of angle AOB is θ radians

The perimeter of the region R shown shaded in Figure 1 is $\frac{28\pi}{9}$ cm

(a) Find the exact value of θ **(2)**

(b) Find the area, in cm^2 to 3 significant figures, of the region R **(4)**

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5. Question 3 · part b · 4 marks

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Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

5. Question 3 · part b · 4 marks · Working and answer

Working and answer

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6. Question 3 · part whole · 6 marks

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Further practice · Attempt part whole only.

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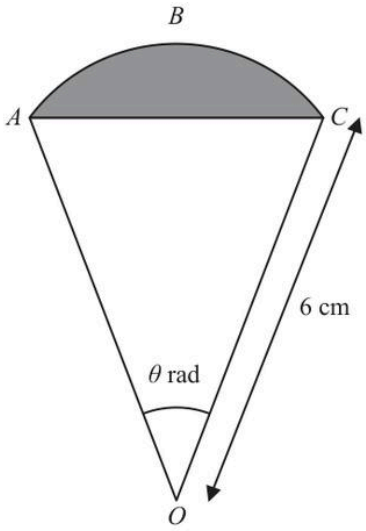


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector $OABC$ of a circle with centre O and radius 6 cm
The size of angle AOC is θ radians.

The perimeter of sector $OABC$ is $(12+\pi)$ cm

The region shown shaded in Figure 1 is bounded by the arc ABC and the line AC
The area of this region is S cm^2

Find the exact value of S

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6. Question 3 · part whole · 6 marks

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Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 6 marks)

6. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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7. Question 4 · part a · 4 marks

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Further practice · Attempt part a only.

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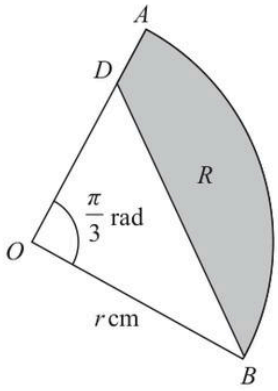


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows sector OAB of a circle with centre O and radius r cm.

The angle $AOB = \frac{\pi}{3}$ radians.

The point D divides OA in the ratio $3 : 1$

The area of the region R, shown shaded in Figure 1, is $79.5\,\text{cm}^2$

(a) Calculate the value, to 2 significant figures, of r

(4)

(b) Calculate the perimeter, in cm to 2 significant figures, of the region R

(4)

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7. Question 4 · part a · 4 marks · Working and answer

Working and answer

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8. Question 4 · part b · 4 marks

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Further practice · Attempt part b only.

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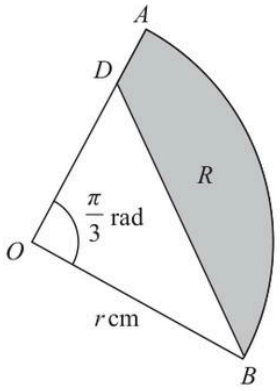


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8. Question 4 · part b · 4 marks

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Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 8 marks)

8. Question 4 · part b · 4 marks · Working and answer

Working and answer

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